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EVERY $K_{2,n}$ IS CONTRACTIBLE

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ABSTRACT. We prove that $K_{2,n}$ is contractible for every $n \geq 1$: every graph containing a $K_{2,n}$ -scheme contains a rooted $K_{2,n}$ minor. This answers the $K_{2,4}$ part of Question 8.2 of Kündgen, Pelsmajer and Ramamurthi. Since contractibility is inherited by subgraphs, every bipartite graph with one part of order at most two is also contractible. The proof encodes the two distinguished colour classes as connected multigraphs on a common labelled edge set. A component count and Edmonds' matroid union theorem yield disjoint spanning trees.

1. INTRODUCTION

Let G and H be graphs with $V(H) \subseteq V(G)$. An H -scheme in G is a family

$$(P_{uv} : uv \in E(H))$$

of paths such that P_{uv} is a u - v path containing no other vertex of H , and every non-empty collection of scheme paths with a common vertex has a common end in H . Their union is the *underlying graph* of the scheme. A *rooted H minor* is a family $(C_v : v \in V(H))$ of pairwise disjoint connected vertex sets such that C_u and C_v are adjacent whenever $uv \in E(H)$, and $v \in C_v$ for every $v \in V(H)$. The graph H is *contractible* if every graph containing an H -scheme contains a rooted H minor.

Kündgen, Pelsmajer and Ramamurthi introduced this terminology and developed the basic theory of contractible graphs [2]. They proved that $K_{2,3}$ is contractible [2, Theorem 6.2], and asked whether $K_{2,4}$ or $K_{3,3}$ is contractible [2, Question 8.2]. We answer the first part and obtain the whole family at once.

Theorem 1.1. *For every integer $n \geq 1$, the graph $K_{2,n}$ is contractible.*

The proof uses the coloured-scheme reduction of Kündgen, Pelsmajer and Ramamurthi and Edmonds' matroid union theorem. The only additional ingredient is the following elementary packing lemma. Auxiliary multigraphs may have parallel edges but no loops; all other graphs are finite and simple.

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2. A TWO-PROJECTION PACKING LEMMA

Lemma 2.1. *Let $m \geq 1$, and let M_1, M_2 be connected multigraphs whose edge sets are both labelled bijectively by the same set E . For $j \in \{1, 2\}$, suppose that M_j is the union of paths $Q_1^j, \dots, Q_m^j$ whose edge sets are pairwise disjoint and have union E . Trivial paths are allowed. Suppose also that these paths all contain a vertex r_j , and every other vertex of M_j belongs to at least two of them. Then the graphic matroids of M_1 and M_2 have disjoint bases. Equivalently, there are disjoint subsets T_1, T_2 of E such that T_j is a spanning tree of M_j .*

Proof. For $X \subseteq E$, let $c_j(X)$ denote the number of components of the spanning subgraph $(V(M_j), X)$, including isolated vertices, and put $Y = E \setminus X$. Count the incidences between these components and the paths Q_i^j . The component containing r_j meets all m paths, while every other component meets at least two. Hence

$$(2.1) \quad \sum_C |\{i : V(Q_i^j) \cap V(C) \neq \emptyset\}| \geq m + 2(c_j(X) - 1).$$

Deleting the edges in Y breaks Q_i^j into at most $|E(Q_i^j) \cap Y| + 1$ pieces. Distinct pieces may lie in the same component of $(V(M_j), X)$, so the left side of (2.1) is at most

$$\sum_{i=1}^m (|E(Q_i^j) \cap Y| + 1) = |Y| + m.$$

It follows that

$$(2.2) \quad c_j(X) \leq \frac{|Y|}{2} + 1.$$

Let $\mathcal{M}_j$ be the graphic matroid of M_j , with rank function ρ_j . Since M_j is connected,

$$\rho_j(E) = |V(M_j)| - 1, \quad \rho_j(X) = |V(M_j)| - c_j(X).$$

Adding (2.2) for $j = 1, 2$ gives

$$(2.3) \quad |E \setminus X| + \rho_1(X) + \rho_2(X) \geq \rho_1(E) + \rho_2(E)$$

for every $X \subseteq E$. Edmonds' matroid union formula [1] says that

$$\rho_{\mathcal{M}_1 \vee \mathcal{M}_2}(E) = \min_{X \subseteq E} (|E \setminus X| + \rho_1(X) + \rho_2(X)).$$

By (2.3), this rank is at least $\rho_1(E) + \rho_2(E)$; taking $X = E$ in the displayed minimum gives the reverse inequality. Choose a maximum independent set $I = I_1 \cup I_2$ of the union, where I_j is independent in $\mathcal{M}_j$. Its cardinality is $\rho_1(E) + \rho_2(E)$. Equality in

$$|I_1 \cup I_2| \leq |I_1| + |I_2| \leq \rho_1(E) + \rho_2(E)$$

shows that I_1 and I_2 are disjoint bases. Taking $T_j = I_j$ proves the lemma. $\square$

3. PROOF OF THE THEOREM

Kündgen, Pelsmajer and Ramamurthi proved that an H -scheme can be reduced, by root-preserving minor operations, to a *coloured H -scheme* [2, Lemma 3.3]. Such a scheme has a proper colouring

$$f : V(G) \longrightarrow V(H)$$

which fixes $V(H)$, and P_{uv} uses only the colours u and v . Moreover, every vertex outside $V(H)$ has degree at least four. We shall also use the following consequences recorded in [2, Remark 3.2(1), (2), and (7)]: colours alternate on every scheme path; every edge lies on exactly one scheme path; and, when u has degree two in H , every vertex of colour u lies on both scheme paths incident with u .

Proof of Theorem 1.1. Let the two vertices in the part of order two be a, b , and let the other vertices be $t_1, \dots, t_n$. By the coloured-scheme reduction, it is enough to find the required rooted minor in the underlying graph G of a coloured $K_{2,n}$ -scheme. Write

$$A = f^{-1}(a), \quad B = f^{-1}(b), \quad L_i = f^{-1}(t_i),$$

and set

$$(3.1) \quad E_i = L_i \setminus \{t_i\}, \quad E = \bigcup_{i=1}^n E_i.$$

Every $x \in E_i$ lies internally on both P_{at_i} and P_{bt_i} . Thus x has two distinct neighbours in A on the first path and two distinct neighbours in B on the second. Form an abstract edge-labelled multigraph M_a on vertex set A by replacing each $x \in E$ with an edge, labelled x , between its two neighbours in A . Define M_b on B in the same way, using the same labelled edge set E .

Suppressing the vertices of E_i on P_{at_i} gives a path Q_i^a in M_a , from a to the A -neighbour of t_i . If $P_{at_i} = at_i$, this is the trivial path at a . Alternation gives

$$(3.2) \quad E(Q_i^a) = E_i.$$

Thus the edge sets of $Q_1^a, \dots, Q_n^a$ are pairwise disjoint and have union E . Every vertex of $A \setminus \{a\}$ lies on a scheme path incident with a , so these paths also cover $V(M_a)$. Their union is therefore M_a , which is connected. A vertex in $A \setminus \{a\}$ lying on k of the paths has degree $2k$ in G : it is internal on each path, and distinct scheme paths share no edge. Since its degree is at least four, $k \geq 2$. The same argument on the b side gives paths Q_i^b with $E(Q_i^b) = E_i$, and every vertex of $B \setminus \{b\}$ belongs to at least two of them.

Lemma 2.1 now gives disjoint sets $T_a, T_b \subseteq E$ such that T_a is a spanning tree of M_a and T_b is a spanning tree of M_b . In G , put

$$(3.3) \quad \begin{aligned} C_a &= A \cup \{x \in E : x \in T_a\}, \\ C_b &= B \cup \{x \in E : x \in T_b\}, \\ C_i &= \{t_i\} \quad (1 \leq i \leq n). \end{aligned}$$

These sets are pairwise disjoint. An edge of T_a labelled x , with ends u, v , corresponds to the two-edge path with vertex sequence u, x, v in G , so $G[C_a]$ is connected; similarly $G[C_b]$ is connected. The edge of P_{at_i} incident with t_i joins C_i to $A \subseteq C_a$, and the corresponding edge of P_{bt_i} joins C_i to $B \subseteq C_b$. The sets in (3.3) therefore form a rooted $K_{2,n}$ -minor model.

Finally, the coloured scheme lies in a rooted minor of the original scheme graph. Composing the two rooted minor models gives a rooted $K_{2,n}$ minor in the original graph. $\square$

Contractibility is inherited by subgraphs [2, Lemma 2.2]. The theorem therefore has the following immediate consequence.

Corollary 3.1. *Every bipartite graph with a part of order at most two is contractible.*

The argument is computation-free. It does not address the $K_{3,3}$ part of Question 8.2 or the wider question of whether every bipartite graph is contractible.

STATEMENT ON AI USE AND INDEPENDENCE

GPT-5.6 Sol, an OpenAI model, contributed to theorem discovery, proofs, verification and drafting. Cavit Erginsoy directed the work and made the final decisions. Erginsoy is not affiliated with OpenAI, and OpenAI neither sponsored nor verified the work.

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